

JHELUM, PakistanWMO: **415980**Lat: **32.933N**Lon: **73.717E**Elev: **768**StdP: **14.29**Time Zone: **5.00 (E05)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 38.5	(c) 40.9	(d) 30.3	(e) 25.1	(f) 53.9	(g) 33.4	(h) 28.8	(i) 51.3	(j) 8.1	(k) 57.9	(l) 7.1	(m) 60.2	(n) 0.1	(o) N/A	(p) 0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	21.8	108.6	74.4	105.5	74.1	102.5	74.7	84.7	95.3	84.0	94.0	83.2	92.9	3.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.4	173.9	91.5	81.6	169.0	90.8	80.8	164.4	90.2	50.2	94.8	49.2	93.8	48.2	93.1	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.3	(b) 7.7	(c) 6.1	(e) 34.3	(f) 113.5	(g) 3.8	(h) 2.3	(i) 31.6	(j) 115.1	(k) 29.3	(l) 116.4	(m) 27.2	(n) 117.7	(o) 24.4	(p) 119.4		
(4) 10.3	(5) 7.7	(6) 6.1	(7) 34.1	(8) 87.0	(9) 2.9	(10) 1.2	(11) 32.0	(12) 87.9	(13) 30.3	(14) 88.6	(15) 28.7	(16) 89.3	(17) 26.6	(18) 90.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.1	53.8	60.2	69.1	79.8	87.7	90.9	87.9	86.5	84.4	77.4	65.9	56.5
	DBStd	13.56	4.14	4.80	5.83	5.27	4.93	5.03	4.08	3.45	3.34	4.78	4.02	3.95
	HDD50	17	12	1	0	0	0	0	0	0	0	0	0	4
	HDD65	824	348	147	28	0	0	0	0	0	0	0	36	265
	CDD50	9166	130	287	592	894	1169	1227	1175	1131	1032	848	477	204
	CDD65	4498	1	13	155	444	704	777	710	666	582	383	63	0
	CDH74	59967	9	139	1431	5137	10287	12216	10338	9156	7023	3643	574	14
	CDH80	34225	0	10	508	2841	6545	8119	6029	4936	3495	1645	97	0
Wind	WSAvg	1.4	0.9	1.4	1.6	1.9	2.1	2.3	1.7	1.4	1.1	0.8	0.6	0.7
Precipitation	PrecAvg	32.4	1.3	1.9	1.8	1.5	1.1	2.4	8.9	8.7	3.0	0.6	0.4	0.9
	PrecMax	48.8	4.0	6.3	6.7	5.3	3.3	7.5	19.7	21.5	9.9	3.3	2.0	4.7
	PrecMin	17.8	0.0	0.0	0.0	0.0	0.0	0.2	2.5	1.6	0.0	0.0	0.0	0.0
	PrecStd	7.3	1.1	1.6	1.7	1.3	0.8	1.8	4.1	3.8	2.5	0.8	0.5	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.5	81.5	94.4	103.9	111.3	112.6	106.7	99.4	98.1	95.5	85.6	76.0
		MCWB	58.0	62.2	67.3	70.5	72.5	73.9	80.5	83.5	80.9	76.8	66.2	59.8
	2%	DB	70.3	78.0	89.3	100.4	107.8	109.7	102.1	97.7	96.3	93.4	82.9	72.6
		MCWB	55.6	59.6	67.9	69.7	73.1	74.4	81.8	83.5	80.1	75.6	65.7	58.5
	5%	DB	67.6	74.9	85.6	97.0	104.8	107.1	99.1	96.2	94.7	91.4	80.4	70.1
		MCWB	54.4	58.8	66.3	68.9	72.6	75.2	81.7	83.0	79.7	74.2	64.3	57.5
	10%	DB	64.6	71.5	81.9	93.6	101.7	104.0	96.7	94.5	93.0	88.8	77.5	67.5
		MCWB	53.4	57.6	64.3	68.7	72.2	75.3	81.8	82.4	79.1	72.5	63.2	56.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.8	65.2	73.1	76.5	80.2	84.1	86.0	85.5	84.6	80.9	70.5	62.6
		MCDB	69.8	77.1	87.1	94.1	100.3	98.4	98.6	95.7	95.0	92.3	81.5	72.1
	2%	WB	57.7	62.6	70.1	73.6	77.5	82.5	84.7	84.8	82.8	78.6	68.5	60.5
		MCDB	67.1	74.2	85.1	93.3	98.1	96.6	95.8	94.6	92.7	89.4	79.3	70.5
	5%	WB	56.1	60.7	68.1	71.8	76.0	81.3	83.8	84.1	81.8	77.0	66.5	58.6
		MCDB	64.7	71.4	81.3	91.1	96.9	95.5	94.2	93.6	91.4	86.9	77.6	68.0
	10%	WB	54.4	58.8	66.1	70.3	74.6	80.0	83.1	83.4	80.8	75.2	64.5	56.7
		MCDB	62.4	68.4	78.7	88.8	95.1	94.4	93.1	92.2	90.0	85.6	74.7	65.6
Mean Daily Temperature Range	5% DB	MDBR	21.1	20.7	22.0	25.2	25.5	21.8	14.8	13.8	16.9	22.8	24.4	23.2
		MCDBR	27.0	26.1	27.4	30.2	30.0	27.0	18.5	16.0	18.8	23.3	27.2	26.8
		MCWBR	14.4	11.5	10.6	9.7	9.2	7.8	6.6	6.5	7.2	9.3	13.1	14.8
	5% WB	MCDBR	23.0	21.8	24.9	27.8	26.6	19.1	15.4	15.1	16.4	20.6	23.9	25.1
		MCWBR	13.4	11.0	11.4	10.6	10.8	8.0	6.6	6.7	7.4	9.3	13.1	15.0
Clear-Sky Solar Irradiance	taub	0.456	0.497	0.506	0.543	0.588	0.679	0.738	0.686	0.591	0.570	0.524	0.470	
	taud	1.939	1.893	1.902	1.858	1.779	1.641	1.586	1.681	1.828	1.769	1.806	1.915	
	Ebn at Noon	228	234	245	242	233	212	198	206	219	209	203	214	
	Edh at Noon	48	56	59	65	70	81	85	76	63	62	55	47	
All-Sky Solar Radiation	RadAvg	924	1151	1578	1936	2224	2181	1824	1699	1688	1502	1133	937	
	RadStd	103	148	142	107	93	117	143	126	93	74	60	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page